Pre-lab Name: Colby Alley

Lab Partner Alek Presley

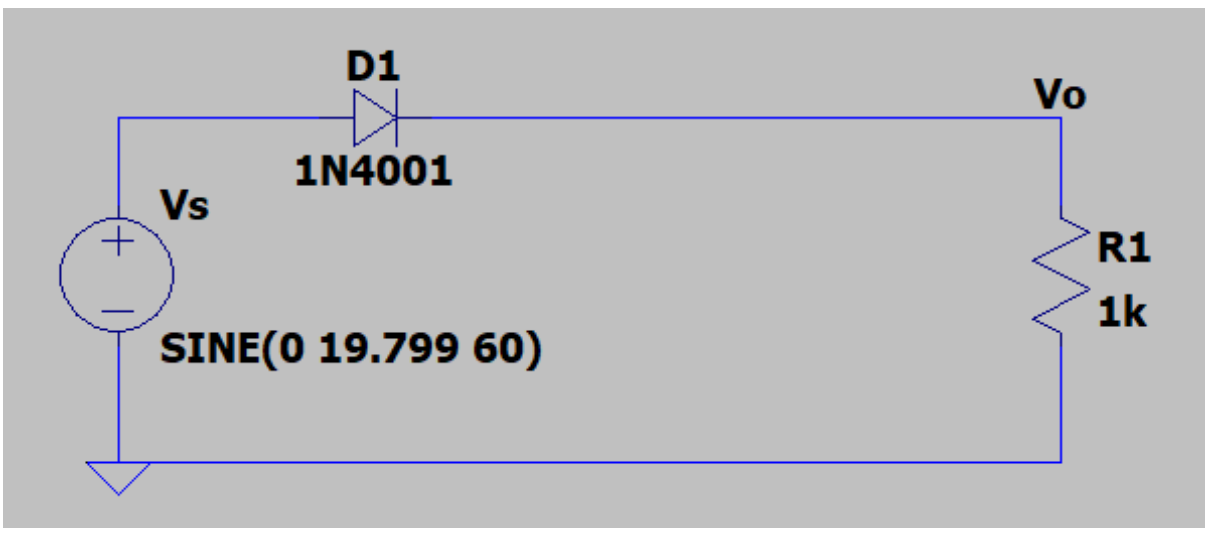
LTspice for Pre-lab

(After this document there is regular computations for pre-lab.)

LTspice diode 1N4001 directive below:

.model 1N4001 D(Is=14.11n N=1.984 Rs=33.89m Ikf=94.81 Xti=3 Eg=1.11 Cjo=25.89p M=.44+ Vj=.3245 Fc=.5 Bv=75 Ibv=10u Tt=5.7u Iave=1 Vpk=50mfg=GI type=silicon)

Circuit 1 (without filter capacitor):



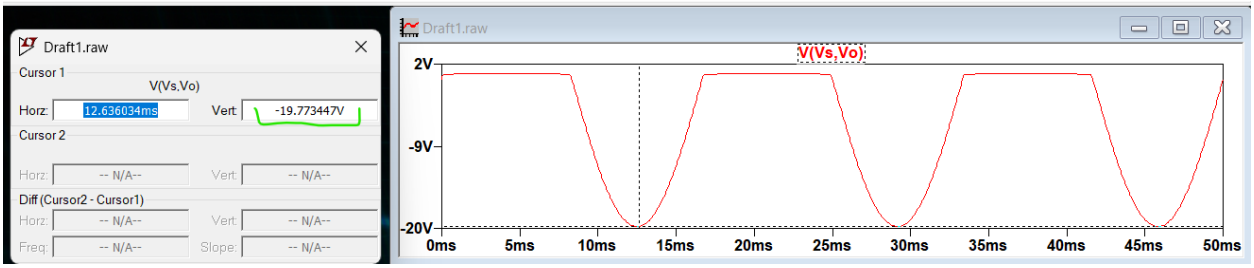
Derived peak amplitude (V_m): 19.779 V, load resistor is 1 kΩ here and diode is 1N4001.

Frequency is 60 Hz.

Value verification (waveforms) from computations on old pre-lab:

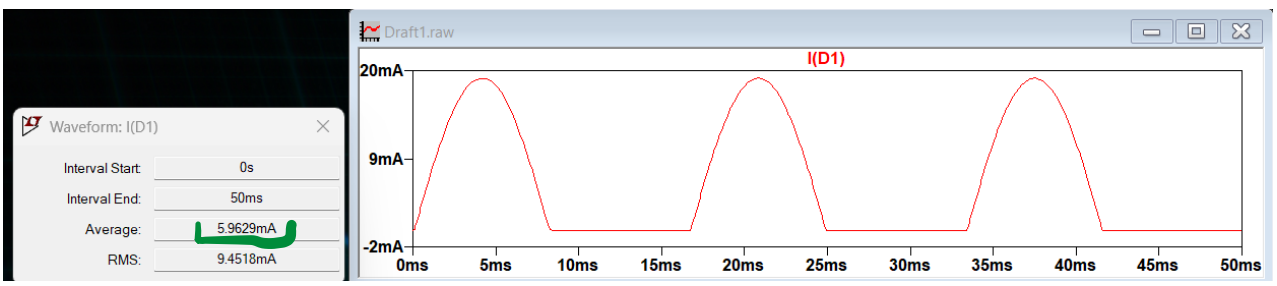
For PL1

For V_{RRM}:



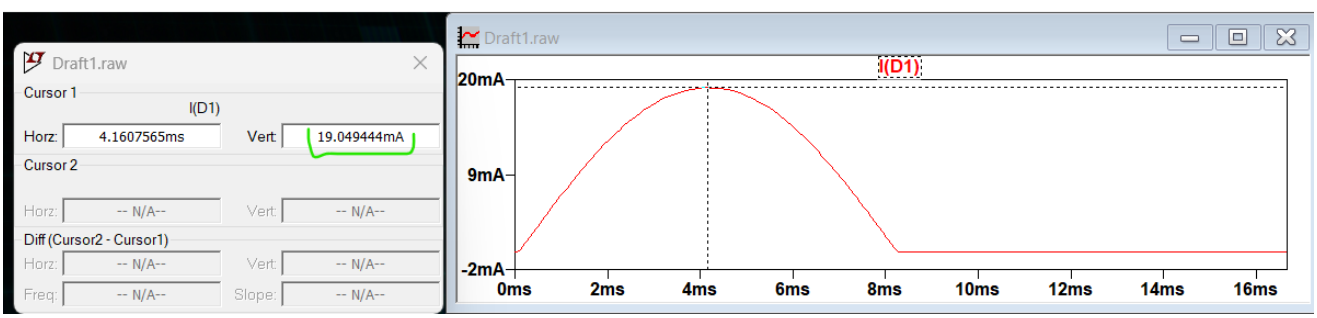
Since $V_{RRM} = 50.0$ V from the data sheet and measured is around -19.773 V, the diode will not exceed limit rating. This shares roughly 3 periods for verification.

For I_O:



Since $I_O = 1.0$ A, and the simulation states 5.963 mA, the diode will not exceed the limit rating. Shows 1 period to verify I_O. This shares roughly 3 periods for verification.

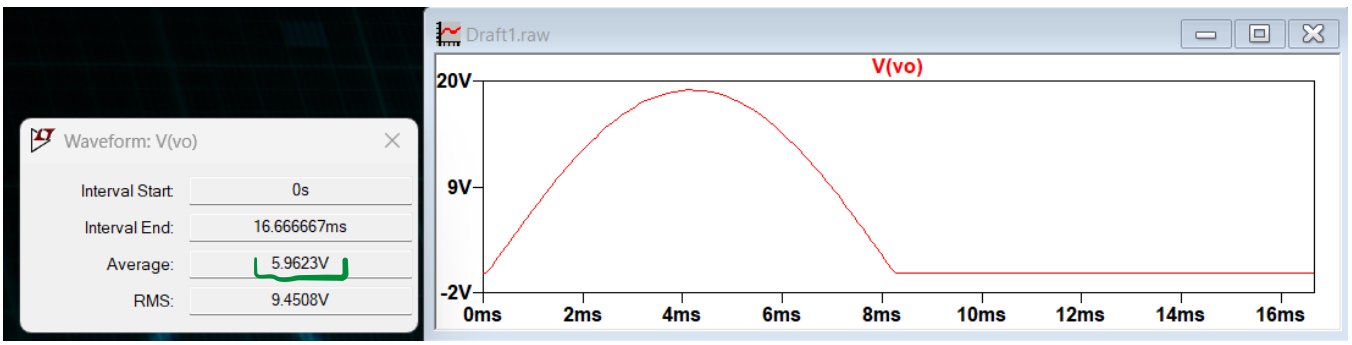
For I_{FSM}:



Since $I_{FSM} = 30.0$ A, and the simulation states 19.103 mA, the diode will not exceed the limit rating. Shows 1 period to verify I_{FSM}.

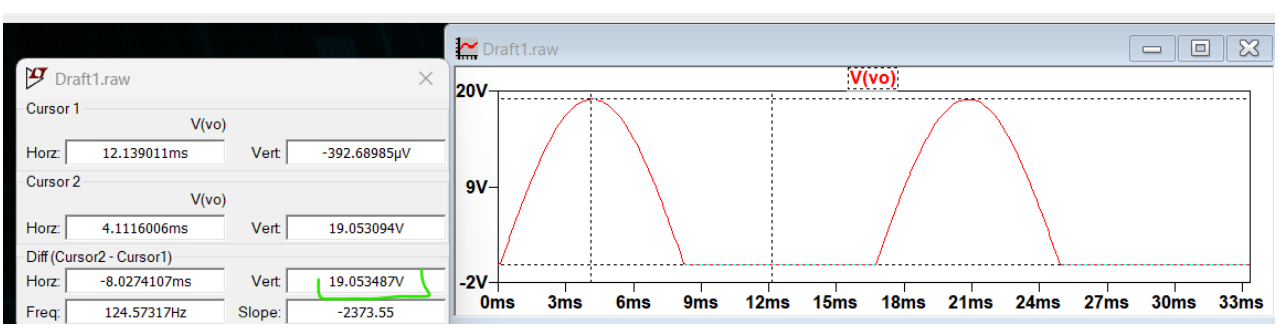
For PL2

For V_{DC}:



$V_{DC} = 6.079$ V (computed) and LTspice says $V_{DC} \approx 5.962$ V. Values are very similar. Shows 1 period to verify D.C. voltage component.

For V_R:



$V_R = 19.099$ V (computed) and LTspice says $V_R \approx 19.053$ V. Values are very similar. Shows 2 periods to verify ripple voltage.

Circuit 2 (without filter capacitor):

In LTspice, to couple inductors, use the directive "K L1 L2 [numerical factor]," where the factor determines the transformer efficiency. In this lab, use the value 1. For some simplicity, let $L_1 = 100$ mH so L_2 will be solved for.

(170 V_p comes from 120 V_{rms} from wall outlet)

L (inductance is proportional to N^2) so:

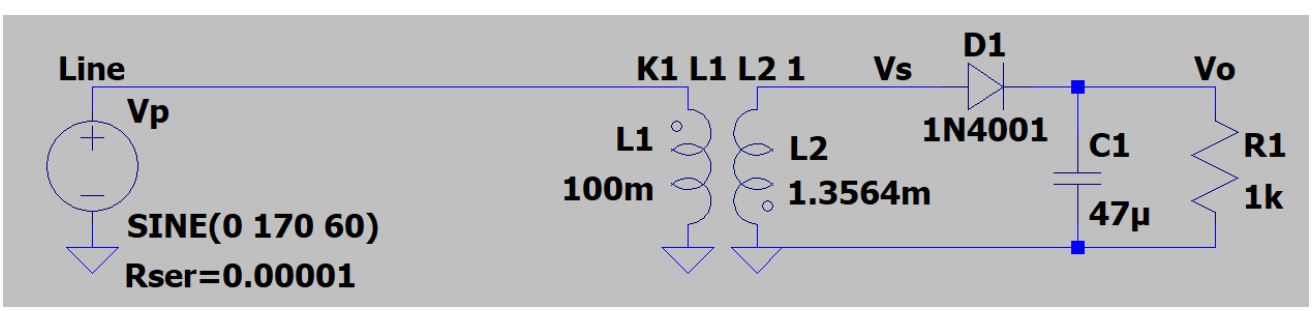
$$\frac{N_1}{N_2} = \sqrt{\frac{L_1}{L_2}}$$

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} \rightarrow \frac{170\ V_p}{19.799\ V_p} \rightarrow \frac{N_1}{N_2} = 8.5863$$

As said above, $L_1 = 100$ mH and the equation $\frac{N_1}{N_2} = \sqrt{\frac{L_1}{L_2}}$ stands. Solving for L_2…

$$L_2 = (L_1)\left(\frac{N_2}{N_1}\right)^2 \rightarrow (100\text{x}10^{-3})\left(\frac{1}{8.5863}\right)^2 = 1.3564\text{ mH}$$

Using these values in LTspice yields the correct transformer winding below:

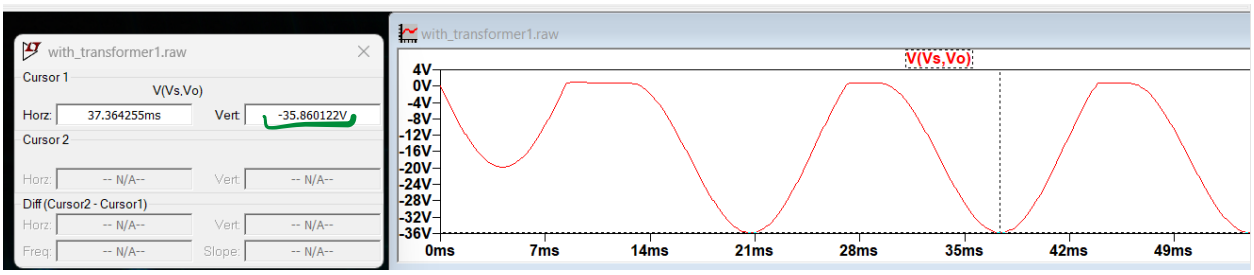


Derived peak amplitude (V_m): 170 V (for the primary wall outlet), load resistor is 1 kΩ here and diode is 1N4001 as well as the capacitor is 47 μF. Frequency is 60 Hz. The parasitic resistance (R_{ser}) of 0.00001 Ω is needed so LTspice runs appropriately.

Value verification (waveforms) from computations on old pre-lab:

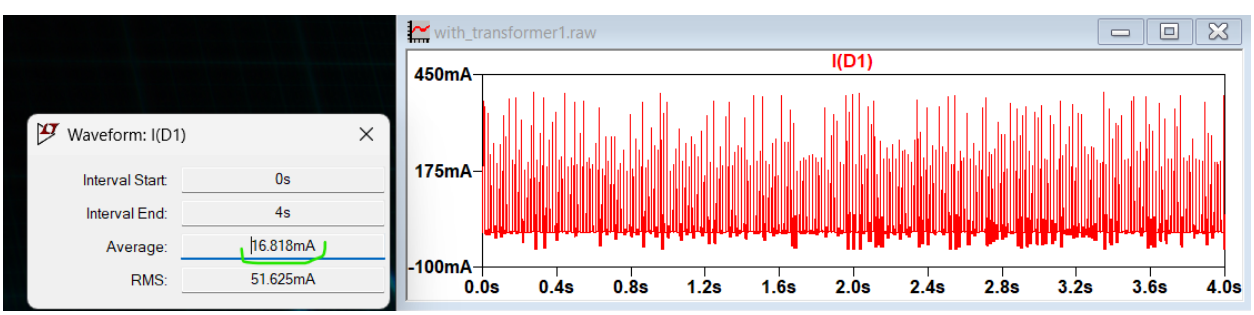
For PL1

For V_{RRM}:



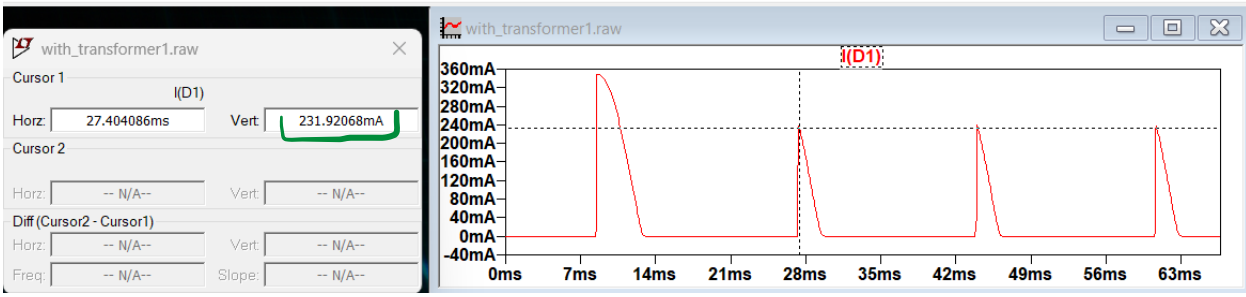
Since $V_{RRM} = 50.0$ V from the data sheet and measured is around -35.860 V (after the capacitor has reached steady state), the diode will not exceed limit rating. This shares 4 periods for verification.

For I_O:



Since $I_O = 1.0$ A, and the simulation states 16.818 mA, the diode will not exceed the limit rating. Shows 240 periods (4 seconds) to verify I_O.

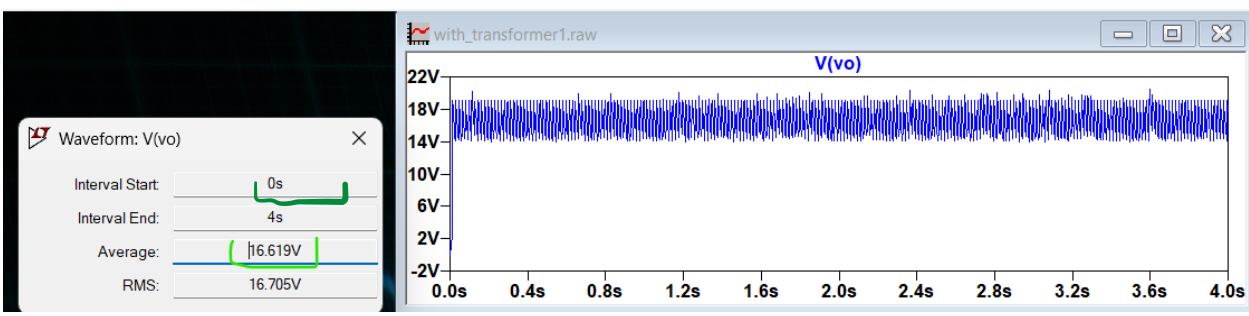
For I_{FSM}:



Since $I_{FSM} = 30.0$ A, and the simulation states 231.921 mA (after the capacitor has reached steady state), the diode will not exceed the limit rating. Shows 4 periods to verify I_{FSM}.

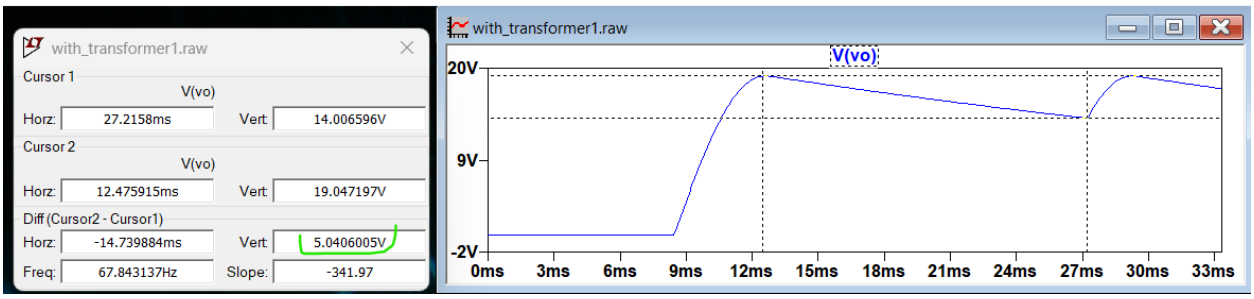
For PL2

For V_{DC}:



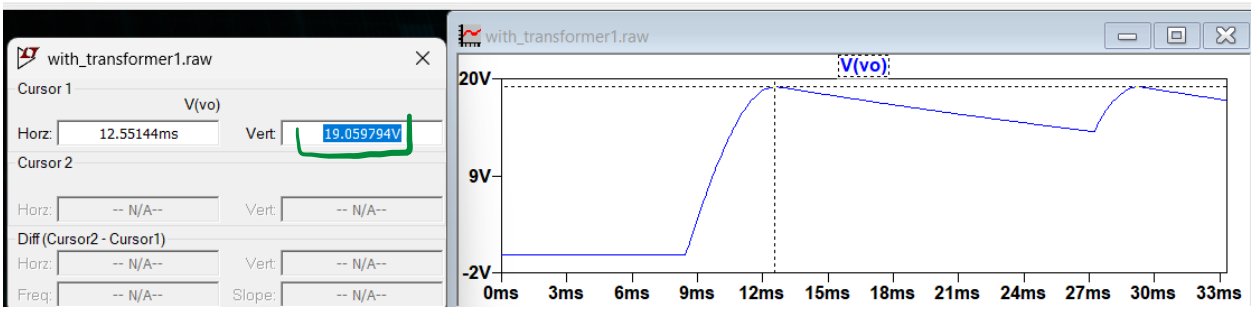
$V_{DC} = 15.713$ V (computed) and LTspice says $V_{DC} \approx 16.619$ V. Values are very similar. Shows 240 periods (4 seconds) to verify D.C. voltage component. Less periods means less accuracy in measurement.

For V_R:



$V_R = 6.773$ V (computed) and LTspice says $V_R \approx 5.047$ V. Values are very similar. It shows 2 periods to verify ripple voltage. LTspice appears to be more accurate. The lab equation is more of an approximation.

For V_C:



$V_C = 19.099$ V (computed) and LTspice says $V_C \approx 19.060$ V. Values are very similar. It shows 2 periods to verify capacitor peak voltage.

*End of fixed pre-lab as everything else is correct.